

Solution of the tutorial on non-parametric estimation TSIA202b

Roland Badeau

Exercise 1

1. $\mathbb{E} \left(Y(v_k) \overline{Y(v_r)} \right) = \mathbb{E} \left(\frac{1}{\sqrt{N}} \sum_{t=0}^{N-1} y_t e^{-2i\pi v_k t} \frac{1}{\sqrt{N}} \sum_{t'=0}^{N-1} y_{t'} e^{2i\pi v_r t'} \right) = \frac{1}{N} \sum_{t=0}^{N-1} \sum_{t'=0}^{N-1} \mathbb{E}(y_t y_{t'}) e^{2i\pi(v_r t' - v_k t)}$
 - if $t \neq t'$ then, since y is white noise (independence + zero mean): $\mathbb{E}(y_t y_{t'}) = \mathbb{E}(y_t) \mathbb{E}(y_{t'}) = 0$
 - if $t = t'$ then : $\mathbb{E}(y_t y_t) = \sigma^2$.

Therefore:

- if $v_k \neq v_r$: $\mathbb{E} \left(Y(v_k) \overline{Y(v_r)} \right) = \frac{\sigma^2}{N} \sum_{t=0}^{N-1} e^{2i\pi(v_r - v_k)t} = \frac{\sigma^2}{N} \times \frac{1 - e^{2i\pi(v_r - v_k)N}}{1 - e^{2i\pi(v_r - v_k)}} = 0$ because $(v_r - v_k)N \in \mathbb{Z}$.
- if $v_k = v_r$: $\mathbb{E} \left(Y(v_k) \overline{Y(v_k)} \right) = \frac{\sigma^2}{N} \times N = \sigma^2$.

Conclusion:

$$\mathbb{E} \left(Y(v_k) \overline{Y(v_r)} \right) = \begin{cases} 0 & \text{if } k \neq r \\ \sigma^2 & \text{else} \end{cases}$$

2. We must prove that: $\mathbb{E}(\hat{S}_{P,XX}(v_k)) = \sum_{l=-\infty}^{\infty} \mathbb{E}(y_l \overline{y_{l-l}}) e^{-2i\pi v_k l}$. According to the previous question, we have:

$$\mathbb{E}(\hat{S}_{P,XX}(v_k)) = \mathbb{E}(|Y(v_k)|^2) = \mathbb{E}(Y(v_k) \overline{Y(v_k)}) = \sigma^2.$$

However, all the terms of the PSD are zero except for $l = 0$. Therefore:

$$\sum_{l=-\infty}^{\infty} \mathbb{E}(y_l \overline{y_{l-l}}) e^{-2i\pi v_k l} = \sigma^2 e^0 = \sigma^2.$$

This proves that the proposed estimator is unbiased.

3. The transform (or rather the product) $Y(v)Y(v)^*$ is a polynomial of degree $N - 1$ in $\cos(2\pi v)$ and this should also be the case of its mathematical expectation. As the estimate is unbiased for N frequencies v_k , it becomes so for all v , all the more that it is a white noise whose PSD is constant.

Exercise 2

1.

$$\begin{aligned}
 MSE(\hat{y}) &= \mathbb{E}((\hat{y} - y)^2) = \mathbb{E}((\hat{y} - \mathbb{E}(\hat{y}) + \mathbb{E}(\hat{y}) - y)^2) = \mathbb{E}((\hat{y} - \mathbb{E}(\hat{y}))^2 + 2(\hat{y} - \mathbb{E}(\hat{y}))(\mathbb{E}(\hat{y}) - y) + (\mathbb{E}(\hat{y}) - y)^2) \\
 &= \mathbb{E}((\hat{y} - \mathbb{E}(\hat{y}))^2) + 2\mathbb{E}[(\hat{y} - \mathbb{E}(\hat{y}))](\mathbb{E}(\hat{y}) - y) + (\mathbb{E}(\hat{y}) - y)^2 \\
 &= \mathbb{E}((\hat{y} - \mathbb{E}(\hat{y}))^2) + 2(\mathbb{E}(\hat{y}) - \mathbb{E}(\hat{y}))(\mathbb{E}(\hat{y}) - y) + (\mathbb{E}(\hat{y}) - y)^2 \\
 &= \mathbb{E}((\hat{y} - \mathbb{E}(\hat{y}))^2) + (\mathbb{E}(\hat{y}) - y)^2
 \end{aligned}$$

We have just proved that the mean squared error of an estimator $\hat{y}$ is equal to the sum of its variance and squared bias.

2. We have from question 1 (and from the hypothesis that $\hat{x}$ is unbiased) that:

$$\begin{aligned}
 MSE(\tilde{x}) &= \mathbb{E}((\tilde{x} - \mathbb{E}(\tilde{x}))^2) + (\mathbb{E}(\tilde{x}) - x)^2 = \rho^2 \mathbb{E}((\hat{x} - x)^2) + (\rho - 1)^2 x^2 \\
 &= \rho^2 \sigma_{\hat{x}}^2 + (\rho - 1)^2 x^2
 \end{aligned}$$

If we consider the MSE as a function of ρ , it is a parabola of the form $a\rho^2 + b\rho + c$ with $a > 0$. So the problem is convex and admits a unique minimum corresponding to the zeroing of the derivative of $MSE(\tilde{x})$ w.r.t. ρ :

$$2\rho\sigma_{\hat{x}}^2 + (2\rho - 2)x^2 = 0 \implies \rho = \frac{x^2}{\sigma_{\hat{x}}^2 + x^2}$$

3. By simply taking the definition of ρ , one observes that the minimum is reached for:

$$\tilde{S}_{P,XX}(\nu) = \rho \hat{S}_{P,XX}(\nu)$$

with

$$\rho = \frac{S_{XX}(\nu)^2}{S_{XX}(\nu)^2 + \lim_{N \rightarrow +\infty} \mathbb{E}(\hat{S}_{P,XX}^2(\nu))} = \frac{S_{XX}(\nu)^2}{S_{XX}(\nu)^2 + S_{XX}(\nu)^2} = \frac{1}{2}$$

By using question 1 and the previous answer, we directly get that:

$$\begin{aligned}
 \mathbb{E}((\tilde{S}_{P,XX}(\nu) - S_{XX}(\nu))^2) &= \left(\frac{1}{2} \mathbb{E}(\hat{S}_{P,XX}(\nu)) - S_{XX}(\nu) \right)^2 + \mathbb{E}\left(\frac{\hat{S}_{P,XX}(\nu)^2}{4} \right) \\
 &\xrightarrow{N \rightarrow \infty} \frac{1}{4} S_{XX}(\nu)^2 + \frac{1}{4} S_{XX}(\nu)^2 = \frac{1}{2} \lim_{N \rightarrow \infty} \left[\mathbb{E}((\hat{S}_{P,XX}(\nu) - S_{XX}(\nu))^2) \right]
 \end{aligned}$$

This is a practical method for obtaining an estimator of S_{XX} , biased but halving the MSE. We just have to compute the periodogram $\hat{S}_{P,XX}$ and to divide it by two to get a better estimate of the PSD (in the sense of the reduction of its MSE).

Exercise 3

1. We have to derive $p_{S_{XX}}(\hat{S}_{P,XX})$ w.r.t. S_{XX} and to determine the value that cancels the latter:

$$\begin{aligned}\frac{\partial p}{\partial S_{XX}}(\hat{S}_{P,XX}) &= -\frac{1}{\sqrt{2\pi}S_{XX}^2} e^{-\left(\frac{(\hat{S}_{P,XX}-S_{XX})^2}{2S_{XX}^2}\right)} + \frac{1}{\sqrt{2\pi}S_{XX}} \left(-\frac{1}{2} \times 2 \left(\frac{\hat{S}_{P,XX}}{S_{XX}} - 1\right) \times \left(-\frac{\hat{S}_{P,XX}}{S_{XX}^2}\right)\right) e^{-\left(\frac{(\hat{S}_{P,XX}-S_{XX})^2}{2S_{XX}^2}\right)} \\ &= -\frac{1}{\sqrt{2\pi}S_{XX}^2} e^{-\left(\frac{(\hat{S}_{P,XX}-S_{XX})^2}{2S_{XX}^2}\right)} \left(1 - \left(\frac{\hat{S}_{P,XX}}{S_{XX}} - 1\right) \times \frac{\hat{S}_{P,XX}}{S_{XX}}\right) \\ &= -\frac{1}{\sqrt{2\pi}S_{XX}^2} e^{-\left(\frac{(\hat{S}_{P,XX}-S_{XX})^2}{2S_{XX}^2}\right)} \left(1 - \frac{\hat{S}_{P,XX}^2}{S_{XX}^2} + \frac{\hat{S}_{P,XX}}{S_{XX}}\right)\end{aligned}$$

Let's then study $\frac{\partial p}{\partial S_{XX}}(\hat{S}_{P,XX}) = 0$. It follows that:

$$\frac{\partial p}{\partial S_{XX}}(\hat{S}_{P,XX}) = 0 \Leftrightarrow \left(1 - \frac{\hat{S}_{P,XX}^2}{S_{XX}^2} + \frac{\hat{S}_{P,XX}}{S_{XX}}\right) = 0 \Leftrightarrow S_{XX}^2 + S_{XX}\hat{S}_{P,XX} - \hat{S}_{P,XX}^2 = 0$$

This is a second degree equation w.r.t. S_{XX} to be solved:

$$\Delta = \hat{S}_{P,XX}^2 + 4\hat{S}_{P,XX}^2 = 5\hat{S}_{P,XX}^2 > 0$$

we then get the positive solution (because the PSD is nonnegative):

$$\tilde{S}_{P,XX} = \frac{-\hat{S}_{P,XX} + \sqrt{5}\hat{S}_{P,XX}}{2} = \left(\frac{-1 + \sqrt{5}}{2}\right) \hat{S}_{P,XX}$$

2. Let us compare the MSE of the PSD estimators:

$$\begin{aligned}\mathbb{E}\left(\left(\tilde{S}_{P,XX}^{\text{exo } 3} - S_{XX}\right)^2\right) &= \left(\left(\frac{-1 + \sqrt{5}}{2}\right) \mathbb{E}(\hat{S}_{P,XX}) - S_{XX}\right)^2 + \left(\frac{-1 + \sqrt{5}}{2}\right)^2 \mathbb{E}(\hat{S}_{P,XX}^2) \\ &\rightarrow_{N \rightarrow \infty} \left(\frac{-3 + \sqrt{5}}{2}\right)^2 S_{XX}^2 + \left(\frac{-1 + \sqrt{5}}{2}\right)^2 S_{XX}^2 \\ &= (5 - 2\sqrt{5}) \lim_{N \rightarrow \infty} \left[\mathbb{E}\left((\hat{S}_{P,XX} - S_{XX})^2\right)\right] \\ &\simeq 0.53 \lim_{N \rightarrow \infty} \left[\mathbb{E}\left((\hat{S}_{P,XX} - S_{XX})^2\right)\right]\end{aligned}$$

the MSE is slightly higher in the estimator of exercise 3. Now let us compare the biases of both estimators:

$$\begin{aligned}\lim_{N \rightarrow \infty} \mathbb{E}(\tilde{S}_{P,XX}^{\text{exo } 2}) &= \frac{1}{2} S_{XX} \\ \lim_{N \rightarrow \infty} \mathbb{E}(\tilde{S}_{P,XX}^{\text{exo } 3}) &\simeq 0.62 S_{XX}\end{aligned}$$

The estimator proposed in this exercise is therefore less biased than that in exercise 2.